Title: Evaluating Elliptic-curve Signature Protocols Under Realistic Constraints: Implications for Privacy-preserving Machine Learning

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Authors: T. Lindqvist, T. Okafor, I. Haddad, V. Rossi (Digital Futures Research Group, Harlow University)
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Abstract:

Secure key exchange over lossy links remains a critical challenge for public-key infrastructure, particularly in privacy-preserving machine learning. This paper proposes a framework built on elliptic-curve signature protocols that explicitly accounts for ciphertext expansion reduction. We evaluate the framework on the eBACS benchmarking results and compare it against 6 baseline methods. Experimental results show that the proposed approach improves security level in bits by 11.2% while keeping ciphertext overhead within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of public-key infrastructure has changed how researchers and practitioners approach secure key exchange over lossy links, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, elliptic-curve signature protocols have attracted sustained attention as a promising route toward solving secure key exchange over lossy links in realistic settings such as privacy-preserving machine learning. A central difficulty is that improvements in encryption throughput frequently come at the expense of security level in bits, forcing designers to accept trade-offs that are poorly understood. This paper addresses these gaps by studying how elliptic-curve signature protocols interact with lattice-based encryption schemes when applied to secure key exchange over lossy links.

Our work is motivated by the growing demand for privacy-preserving machine learning, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of secure key exchange over lossy links, a practical design built on elliptic-curve signature protocols, and an empirical evaluation on the eBACS benchmarking results. Understanding ciphertext expansion reduction is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Okafor et al. (2009) introduced an early framework based on elliptic-curve signature protocols, reporting a 42.1% improvement in ciphertext overhead over previous baselines. Subsequent work by Chen et al. (2014) extended this line of research to privacy-preserving machine learning, although the evaluation was restricted to small-scale settings. Researchers working on public-key infrastructure have explored lattice-based encryption schemes as an alternative, with Novak et al. (2012) noting that the approach scales poorly beyond 64 components. OConnor et al. (2010) analysed ciphertext expansion reduction and showed that purely heuristic solutions cannot guarantee consistent security level in bits. More recently, Okafor et al. (2010) combined elliptic-curve signature protocols with lattice-based encryption schemes, achieving a 39.9% reduction in overhead on the eBACS benchmarking results.

Several studies have criticised the reliance on synthetic workloads, arguing that results on the eBACS benchmarking results may not generalise to privacy-preserving machine learning. In contrast to these efforts, our work treats secure key exchange over lossy links and ciphertext expansion reduction jointly rather than as

independent concerns. Industry reports corroborate these academic findings, pointing to secure key exchange over lossy links as a persistent obstacle for privacy-preserving machine learning. Despite this body of literature, no prior study has systematically compared elliptic-curve signature protocols and lattice-based encryption schemes under identical experimental conditions.

3. Methodology

We formalise secure key exchange over lossy links as a constrained optimisation task in which the objective is to maximise security level in bits while keeping key generation latency within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on elliptic-curve signature protocols, and a validation layer that incorporates lattice-based encryption schemes. All experiments were conducted on the eBACS benchmarking results, which provides 1200 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies elliptic-curve signature protocols to reduce the effect of ciphertext expansion reduction, followed by a calibration step that stabilises security level in bits. We compare against 6 baseline methods, including a conventional approach built on lattice-based encryption schemes and two recent variants reported in the literature. The proposed design improves security level in bits by 11.2% relative to the strongest baseline and reduces ciphertext overhead by 38.2%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the lattice-based encryption schemes component reduces security level in bits by approximately 5.5%, confirming its contribution. A sensitivity analysis over 128 parameter settings shows that performance remains stable, with variation in key generation latency below 12.8%. Threats to validity include the limited diversity of the eBACS benchmarking results and the possibility that deployment in privacy-preserving machine learning introduces conditions not captured in our setup.

4. Conclusion

This paper investigated secure key exchange over lossy links through the lens of elliptic-curve signature protocols, and demonstrated that a carefully designed pipeline can improve security level in bits by 11.2% on the eBACS benchmarking results. The findings suggest that public-key infrastructure will continue to benefit from closer integration of elliptic-curve signature protocols and lattice-based encryption schemes, particularly in privacy-preserving machine learning. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of ciphertext expansion reduction. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for secure key exchange over lossy links.